

MHAFUZUL ISLAM

Ph.D. | Robotics · Machine Learning · Cybersecurity · Cyber-Physical Systems · Autonomous Vehicles

West Bloomfield, MI | 864-986-5446 | mdmhafi@gmail.com | [Google Scholar](#) | [GitHub](#)

h-index: 16 · Citations: 800+ · 15+ Peer-Reviewed Publications · 18+ U.S. Patents

RESEARCH INTERESTS

Machine Learning and Artificial Intelligence · Connected and Automated Vehicles · Autonomous systems and robotics · Computer Vision · Cyber-physical systems security and resilience · Smart Mobility · Data Analytics · Connected and intelligent transportation systems · Neuromorphic Computing · Bio-inspired Autonomous Systems

EDUCATION

Doctor of Philosophy (Ph.D.), Civil Engineering 2021

Clemson University, Clemson, South Carolina

Dissertation: *Vision-Based Navigation and Cybersecurity for Autonomous Vehicles in Roadway Environments*

Advisor: Prof. Mashrur (Ronnie) Chowdhury, Eugene Douglas Mays Chair of Transportation

Master of Science (M.S.), Civil Engineering 2018

Clemson University, Clemson, South Carolina

Advisor: Prof. Mashrur (Ronnie) Chowdhury, Eugene Douglas Mays Chair of Transportation

Bachelor of Science (B.Sc.), Computer Science and Engineering 2014

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

PROFESSIONAL EXPERIENCE

Senior Researcher — Automotive Sensing March 2025 – Present

Imec USA R&D (Leuven Inc.), Michigan

Deep learning-based perception for autonomous platforms (LiDAR, RGB, SWIR/IR); BEV model development for adverse weather; field data collection at University of Michigan MCity.

Automated Vehicle & Robotics Engineer Aug 2023 – Feb 2025

General Motors Defense, Warren, Michigan

Autonomous vehicle perception pipelines, motion planning, obstacle avoidance, and field deployment on defense ground vehicles in unstructured environments.

Senior Researcher — Autonomous Driving Aug 2021 – Aug 2023

General Motors R&D, Warren, Michigan

Deep learning algorithms for AV perception in extreme environments; simulation validation; Lunar Mobility Vehicle program. 18+ U.S. patents filed. 1 GM Trade Secret awarded.

Machine Learning Scientist Jan 2021 – Aug 2021

MicroVision Inc., Redmond, Washington

3D object detection and tracking; real-time LiDAR-camera sensor fusion in C++/Python.

Graduate Research Assistant Aug 2016 – May 2021

Clemson University, Clemson, South Carolina

Advisor: Prof. Mashrur Chowdhury. Research on AV perception, V2I/V2X cybersecurity, sensor fusion, and DNN-based navigation for connected and autonomous vehicles.

Software Engineer Jan 2014 – May 2016

Infineon Technologies, Munich, Germany

Embedded firmware development (C++) for barometric pressure and microphone sensors on Infineon XMC microcontroller platforms

Led system architecture design and software development of a Smart Home Automation System. Designed and developed firmware algorithms for Smart Meter systems.

PUBLICATIONS

(* denotes corresponding/lead author)

Referred Journal Articles

- [J15] K. H. Shakib, M. Rahman, **M. Islam**, and M. Chowdhury. *Impersonation attack using quantum Shor's algorithm against blockchain-based vehicular ad-hoc network*. IEEE Transactions on Intelligent Transportation Systems, 2025. doi: 10.1109/TITS.2025.3534656
- [J14] **M. Islam***, M. Chowdhury, M. Rahman, and Z. Khan. *Time-to-approach-based model predictive controller for autonomous vehicle safety during unexpected wheelchairs on the path*. Cyber-Physical Systems, pp. 1–22, 2023. doi: 10.1080/23335777.2023.2264347
- [J13] S. Dasgupta, M. Rahman, **M. Islam**, and M. Chowdhury. *A sensor fusion-based GNSS spoofing attack detection framework for autonomous vehicles*. IEEE Transactions on Intelligent Transportation Systems, 2022. doi: 10.1109/TITS.2022.3197817
- [J12] M. Rahman, M. R. **Islam**, **M. Islam**, M. Chowdhury, and T. Khan. *A physics-based longitudinal driver model for automated vehicles*. IEEE Access, 10:80883–80899, 2022. doi: 10.1109/ACCESS.2022.3193997
- [J11] Z. Khan, S. M. Khan, M. Chowdhury, M. Rahman, and **M. Islam**. *Performance evaluation of 5G millimeter-wave-based vehicular communication for connected vehicles*. IEEE Access, 2022. doi: 10.1109/ACCESS.2022.3158669
- [J10] M. S. Salek, S. M. Khan, M. Rahman, H. Deng, **M. Islam**, Z. Khan, M. Chowdhury, and M. Shue. *A review on cybersecurity of cloud computing for supporting connected vehicle applications*. IEEE Internet of Things Journal, 9(11):8250–8268, 2022. doi: 10.1109/JIOT.2022.3152477
- [J9] G. Comert, M. Rahman, **M. Islam**, and M. Chowdhury. *Change point models for real-time cyber attack detection in connected vehicle environment*. IEEE Transactions on Intelligent Transportation Systems, 23(8):12328–12342, 2022. doi: 10.1109/TITS.2021.3113675
- [J8] M. Rahman, **M. Islam**, C. Holt, J. Calhoun, and M. Chowdhury. *Dynamic error-bounded lossy compression to reduce the bandwidth requirement for real-time vision-based pedestrian safety applications*. Journal of Real-Time Image Processing, 19(1):117–131, 2022. doi: 10.1007/S11554-021-01165-0
- [J7] **M. Islam***, M. Rahman, M. Chowdhury, G. Comert, E. D. Sood, and A. Apon. *Vision-based personal safety messages (PSMs) generation for connected vehicles*. IEEE Transactions on Vehicular Technology, 69(9):9402–9416, 2020. doi: 10.1109/TVT.2020.2982189
- [J6] Z. Khan, M. Chowdhury, **M. Islam**, C. Y. Huang, and M. Rahman. *Long short-term memory neural network-based attack detection model for in-vehicle network security*. IEEE Sensors Letters, 4(6), 2020. doi: 10.1109/LSENS.2020.2993522
- [J5] **M. Islam***, M. Rahman, S. M. Khan, M. Chowdhury, and L. Deka. *Development and performance evaluation of a connected vehicle application development platform*. Transportation Research Record, 2674(5):537–552, 2020. doi: 10.1177/0361198120917146
- [J4] M. Rahman, **M. Islam**, J. Calhoun, and M. Chowdhury. *Real-time pedestrian detection approach with an efficient data communication bandwidth strategy*. Transportation Research Record, 2673(6):129–139, 2019. doi: 10.1177/0361198119843255
- [J3] **M. Islam***, M. Chowdhury, H. Li, and H. Hu. *Vision-based navigation of autonomous vehicles in roadway environments with unexpected hazards*. Transportation Research Record, 2673(9):1–14, 2019. doi: 10.1177/0361198119855606
- [J2] M. Chowdhury, M. Rahman, A. Rayamajhi, S. M. Khan, **M. Islam**, Z. Khan, and J. Martin. *Lessons learned from the real-world deployment of a connected vehicle testbed*. Transportation Research Record, 2672(22):10–23, 2018. doi: 10.1177/0361198118799034
- [J1] **M. Islam***, M. Chowdhury, H. Li, and H. Hu. *Cybersecurity attacks in vehicle-to-infrastructure applications and their prevention*. Transportation Research Record, 2672(19):66–78, 2018. doi: 10.1177/0361198118799012

Referred Conference Papers & Proceedings

- [C9] W. Jin, M. Chowdhury, **M. Islam**, and M. Rahman. *Risk-based freeway lane-changing decision-making framework for autonomous vehicles in mixed traffic streams*. Transportation Research Board Annual Meeting (TRB), Washington, D.C., 2022.
- [C8] S. Dasgupta, M. Rahman, **M. Islam**, and M. Chowdhury. *Prediction-based GNSS spoofing attack detection for autonomous vehicles*. Transportation Research Board Annual Meeting (TRB), Washington, D.C., 2021.
- [C7] **M. Islam***, M. Rahman, S. M. Khan, M. Chowdhury, and L. Deka. *Development and performance evaluation of a connected vehicle application development platform (CVDeP)*. Transportation Research Board Annual Meeting (TRB), Washington, D.C., 2020.
- [C6] M. Rahman, **M. Islam**, J. Calhoun, and M. Chowdhury. *Real-time pedestrian detection approach with an efficient data communication bandwidth strategy*. Transportation Research Board Annual Meeting (TRB), Washington, D.C., 2019.
- [C5] G. Comert, M. Rahman, **M. Islam**, and M. Chowdhury. *Change-point models for real-time V2I cyber attack detection in a connected vehicle environment*. Transportation Research Board Annual Meeting (TRB), Washington, D.C., 2019.
- [C4] M. Chowdhury, M. Rahman, A. Rayamajhi, S. M. Khan, **M. Islam**, Z. Khan, and J. Martin. *Lessons learned from the real-world deployment of a connected vehicle testbed*. Transportation Research Board Annual Meeting (TRB), Washington, D.C., 2018.
- [C3] **M. Islam***, F. Li, M. Rahman, S. Islam, M. Chowdhury, and Y. Wang. *Human-centered time headway controller for cooperative adaptive cruise control (CACC)*. Transportation Research Board Annual Meeting (TRB), Washington, D.C., 2018.
- [C2] **M. Islam***, M. Chowdhury, H. Li, and H. Hu. *Vision-based lane detection and unexpected hazard avoidance for autonomous vehicles*. Transportation Research Board Annual Meeting (TRB), Washington, D.C., 2019.
- [C1] **M. Islam***, M. Chowdhury, H. Li, and H. Hu. *Machine learning-based cyberattack detection for connected vehicle applications*. IEEE International Conference on Intelligent Transportation Systems (ITSC), 2017.

Conference Presentations

- [P2] F. Walden, S. Dasgupta, M. Rahman, and **M. Islam**. *Improving the environmental perception of autonomous vehicles using deep learning-based audio classification*. ASCE International Conference on Transportation & Development (ICTD), Austin, TX, June 2023.
- [P1] S. Dasgupta, M. Rahman, **M. Islam**, M. Chowdhury, and S. Jones. *A sensor fusion-based GNSS spoofing attack detection framework for autonomous vehicles*. ASCE International Conference on Transportation & Development (ICTD), Seattle, WA, May–June 2022.

PATENTS

All patents assigned to GM Global Technology Operations LLC · 1 GM Trade Secret

Granted U.S. Patents

- [PT6] Islam, M. M., Rabbani, A., and Swafford, R. L. *Method and System for Authenticating and Securing Vehicular Services*. U.S. Patent No. 12,509,020. GM Global Technology Operations LLC, 2025.
- [PT5] Islam, M. M., Adiththan, A., and Romano, R. *Context-Based Reminder System for a Vehicle*. U.S. Patent No. 12,469,377. GM Global Technology Operations LLC, 2025.
- [PT4] Islam, M. M., Adiththan, A., and Zhang, Y. *System and Method of Imitating Target Vehicle Behavior for Automated Driving*. U.S. Patent No. 12,459,540. GM Global Technology Operations LLC, 2025.
- [PT3] Sethu, R., Adiththan, A., Islam, M. M., Peranandam, P. M., Zolfagharian, A., and Aghababaeyan, Z. *Situational Recommendations and Control*. U.S. Patent No. 12,459,359. GM Global Technology Operations LLC, 2025.
- [PT2] Islam, M. M., Peranandam, P. M., Golm, E. B., and Rueda, M. F. *Child Seat Latching System for a Vehicle*. U.S. Patent No. 12,358,403. GM Global Technology Operations LLC, 2025.
- [PT1] Islam, M. M., Peranandam, P. M., Golm, E. B., and Rueda, M. F. *System and Method of Controlling a Child Seat Latching Assembly for a Vehicle*. U.S. Patent No. 12,358,401. GM Global Technology Operations LLC, 2025.

Patent Applications (Published / Pending)

- [PT17] Sarwar, A. and Islam, M. M. *Determining Horizontal Clearance for a Vehicle*. GM Global Technology Operations LLC, 2025.
- [PT16] Sarwar, A. and Islam, M. M. *Determining Vertical Clearance for a Vehicle*. GM Global Technology Operations LLC, 2025.
- [PT15] Islam, M. M., Adiththan, A., Peranandam, P. M., and Sethu, R. *Method for Following a Lead Vehicle Without Vehicle-to-Vehicle Communication*. GM Global Technology Operations LLC, 2025.
- [PT14] Angelopoulos, A., Giusto, P., Adiththan, A., Islam, M. M., and Sarwar, A. *Evaluation System for Determining a Network Connectivity Performance Road Map*. GM Global Technology Operations LLC, 2025.
- [PT13] Peranandam, P. M., Sachdev, J., and Islam, M. M. *Electric Mobility Device Communication System for a Service Area*. GM Global Technology Operations LLC, 2024.
- [PT12] Islam, M. M., Esfahani, A. E. A., and Sarwar, A. *Traffic Sign Prediction for a Vehicle*. GM Global Technology Operations LLC, 2025.
- [PT11] Angelopoulos, A., Adiththan, A., Islam, M. M., Giusto, P., and Enzel, A. *Allocating Computing Resources for a Vehicle Application*. GM Global Technology Operations LLC, 2025.
- [PT10] Vemuri, V. N. S. V., Giusto, P., Adiththan, A., Angelopoulos, A., Islam, M. M., Sergakis, J., and Macdonald, A. J. *System and Method for Network Resource Optimization*. GM Global Technology Operations LLC, 2025.
- [PT9] Adiththan, A., Islam, M. M., Peranandam, P. M., and Raviram, S. *System and Method for Context Oriented Auto Park Assist*. GM Global Technology Operations LLC, 2025.
- [PT8] Adiththan, A., Islam, M. M., Peranandam, P. M., and Sethu, R. *Misbehavior Detection and Intervention*. GM Global Technology Operations LLC, 2025.
- [PT7] Adiththan, A., Islam, M. M., Peranandam, P. M., Sethu, R., and Sarwar, A. *System and Method for Detecting a Trailer Coupler and Planning a Trajectory*. GM Global Technology Operations LLC, 2025.

FUNDED RESEARCH & PROPOSAL CONTRIBUTIONS

Simu5G Integration into GM R&D Connected & Autonomous Vehicle Simulation

2022 – 2023

General Motors R&D, Warren, Michigan

Principal Investigator · Internal Funding: \$200,000

Led and secured internal research funding to integrate the Simu5G 5G NR/LTE simulator into the GM R&D connected and autonomous vehicle simulation stack. Responsible for full project leadership from proposal through execution. Collaborated with GM R&D China, University of Pisa (Dr. Giovanni Nardini, Dr. Antonio Virdis), and GM engineering counterparts.

Lunar Mobility Vehicle — Autonomous Perception & Navigation System

2021 – 2023

General Motors R&D / Lunar Outpost, Warren, Michigan

Technical Lead · Funded: \$5,000,000 · Industry Partnership

Supported and led technical aspects of the Lunar Mobility Vehicle (LMV) program in collaboration with Lunar Outpost. Developed deep learning-based perception algorithms, sensor fusion pipelines, and autonomous navigation systems for the extreme and unstructured environment of the lunar surface.

Press: GM brings innovative battery tech to lunar rover concept — news.gm.com

Tier 1 UTC: Center for Connected Multimodal Mobility (C2M2) — Proposal Contributor

2016 – 2017

U.S. Department of Transportation / Clemson University

Proposal Technical Writer · Total Grant: ~\$8,000,000 · USDOT

Contributed to the technical writing of the C2M2 UTC proposal, specifically authoring sections on pedestrian detection and safety improvement for pedestrians and autonomous vehicles at intersections — covering vision-based detection algorithms, real-time data communication strategies, and system architectures for pedestrian-vehicle interaction safety in connected and automated vehicle environments.

Principal Investigator: Prof. Mashrur Chowdhury, Glenn Department of Civil Engineering, Clemson University.

INTERNATIONAL ACADEMIC & INDUSTRY COLLABORATIONS

University of Pisa, Italy

Contact: Dr. Giovanni Nardini, Dr. Antonio Virdis

Simu5G Project — 5G network simulation and evaluation for connected vehicle infrastructure. Integration of Simu5G into autonomous vehicle simulation stacks.

University of Ottawa, Canada

Contact: Prof. Lionel Briand

Evaluating and developing reinforcement learning algorithms for autonomous vehicle systems during performance validation and verification.

University of Alabama, USA

Contact: Dr. Mizanur Rahman (Connected and Automated Mobility Lab)

Quantum computing security for blockchain-based VANETs. Co-authored: "Impersonation Attack Using Quantum Shor's Algorithm Against Blockchain-Based Vehicular Ad-Hoc Network," IEEE T-ITS (Impact Factor: 7.9), 2025.

University of Michigan, USA

Contact: Dr. Hun Seok Kim

Multi-sensor fusion for autonomous systems — integrating LiDAR, camera, and radar for robust perception in adverse weather conditions, validated at MCity Vehicle Testing Facility.

Ghent University, Belgium

Contact: Dr. David Van Hamme

Joint research on sensor fusion methodologies for autonomous perception systems, covering algorithm design, benchmarking on multi-modal datasets, and deployment strategies.

HONORS & AWARDS

1st Place — Intel AI Competition, Transportation Category (Global)

2018

Intel Corporation · Worldwide Competition

Designed and field-deployed a real-time ADAS system with lane detection and deep learning-based object detection (MobileNet) on a live vehicle.

Top 37 — Lyft Autonomous Vehicle Perception Challenge (Global)

2019

Lyft · Worldwide Competition

Semantic segmentation for AV camera data; 92% accuracy on vehicle and drivable-area classification using FCN.

General Motors Trade Secret Award

2023

General Motors R&D, Warren, Michigan

Recognized for proprietary autonomous driving technology designated as a GM trade secret.

C2M2 Graduate Research Fellowship

2016 – 2021

U.S. DOT Center for Connected Multimodal Mobility (C2M2), Clemson University

Funded by the U.S. Department of Transportation Tier 1 University Transportation Center (UTC).

Dr. James Edwin Clark Outstanding Graduate Student Fellowship

2019

Clemson University, Clemson, South Carolina

Awarded to recognize outstanding research contributions by a graduate student in the Glenn Department of Civil Engineering.

AWS Machine Learning Scholarship

2020

Amazon Web Services (AWS)

Competitive scholarship awarded to high-performing engineers pursuing advanced machine learning skills.

Intel Edge AI Scholarship

2020

Intel Corporation

Awarded for demonstrated excellence in edge AI and computer vision development.

- 1st Runner-Up — Infineon Application Design Contest** 2015
Infineon Technologies AG, Munich, Germany · International
 Developed an application for measuring wind speed and direction using the Infineon barometric pressure sensor. This project was the 1st Runners up in the Infineon application design contest globally.
- 1st Runner-Up — International Robotics Challenge (IRC), Techfest** 2013
Indian Institute of Technology (IIT) Bombay, India · International
- 1st Place — International Robotics Challenge (IRC)** 2012
Bangladesh · National Competition

RESEARCH PROJECTS

- Multi-Sensor Perception for Adverse Weather AV Navigation** 2025 – Present
Imec USA R&D / University of Michigan MCity
 LiDAR, RGB, and SWIR multi-sensor data collection and BEV-based deep learning model development for reliable AV perception in fog, rain, and snow. Collaboration with MCity Vehicle Testing Facility.
- Autonomous Ground Vehicle Perception & Planning** 2023 – 2025
General Motors Defense / U.S. DoD-Sponsored
 Full autonomy stack for defense vehicles in unstructured off-road environments: real-time perception, motion planning, and obstacle avoidance for production field deployment.
- Context-Aware Bayesian Fusion for Multimodal Object Detection** 2022 – 2023
General Motors R&D
 Developed context-aware Bayesian fusion framework for multi-modal object detection, integrating LiDAR and camera modalities. Targeted for publication at top-tier conference/journal.
- Connected Vehicle Cybersecurity — V2I/V2X Applications** 2016 – 2021
U.S. DOT C2M2 UTC, Clemson University
 ML-based cyberattack detection systems (LSTM, quantum-classical NN, GNSS spoofing detection) for vehicle-to-infrastructure applications in the South Carolina Connected Vehicle Testbed (SC-CVT).
- MPC-Based AV Lane Keeping and Collision Avoidance** 2019 – 2020
Clemson University / Udacity Self-Driving Car Program
 Implemented a Model Predictive Controller (MPC) for lane-center tracking using a bicycle vehicle dynamics model. Extended the MPC to incorporate time-to-collision and 2D object detection and tracking for obstacle avoidance, outperforming traditional PID and distance-based MPC approaches.
- NSF Connected Vehicle Application Platform (CVDeP)** 2017 – 2020
NSF / Clemson University
 Developed an edge-centric cyber-physical platform for connected and automated vehicle applications supporting real-time Forward Collision Warning and Queue Warning systems, deployed on the South Carolina Connected Vehicle Testbed.

TEACHING EXPERIENCE

Graduate Teaching Assistant — Clemson University

- CE 8930: Autonomous Vehicle Systems** Fall 2019
Clemson University, Dept. of Civil Engineering
 Co-developed new graduate course with advisor Dr. Mashrur Chowdhury. Served as TA preparing course material, conducting lab sessions, and grading.
- CE 8930: Data Analytics for Intelligent Transportation Systems** Fall 2017
Clemson University, Dept. of Civil Engineering
 Co-developed new graduate course. TA responsibilities: course material development, lab instruction, student mentorship.
- CE: Traffic Engineering & Simulation Lab** Spring 2016, Spring 2017

Clemson University, Dept. of Civil Engineering

Led graduate-level lab sessions on traffic simulation and connected vehicle applications.

K-12 Outreach & Broadening Participation in STEM

Instructor — Summer Course: AV Design with Webots & CARLA

2019 – 2020

Clemson University / C2M2 UTC · K-12 Students

Designed, led, and taught a summer course for K-12 students on autonomous vehicle software development using the Webots and CARLA simulation platforms. Topics covered: AV sensor stack design, perception pipeline development (camera & LiDAR), simulation-based testing and validation, and motion planning algorithm development and evaluation.

Instructor — C2M2 Coders: Python for K-12 Students

Spring–Summer 2020

C2M2 UTC, Clemson University · Online · sites.google.com/view/c2m2-coders

Designed and taught a 6-module Python programming course targeting K-12 students, developed under the USDOT C2M2 STEM outreach program. Developed all course materials including lecture slides, videos, coding assignments, and a capstone project.

Workshop Instructor — AV Sensor & Software Stack with Webots

2019

Benedict College, Columbia, South Carolina

Conducted a hands-on workshop on autonomous vehicle sensor and software stack design using the Webots robotics simulator. Participants designed and implemented AV perception and control pipelines in simulation, covering sensor integration (LiDAR, camera), obstacle detection, and basic navigation.

Courses Available to Teach

- Autonomous Vehicles and Robotics
- Deep Learning and Computer Vision for Robotics
- Multi-Sensor Fusion and State Estimation
- Motion Planning and Control for Autonomous Systems
- Embedded Systems and Real-Time AI
- Transportation Cyber-Physical Systems Security
- Connected and Intelligent Transportation Systems
- Introduction to Machine Learning (undergraduate)

STUDENT MENTORSHIP

Summer Research Interns — General Motors R&D, Warren, Michigan:

- Amirhossein Zolfagharian (Graduate Intern) — co-inventor on U.S. Patent No. 12,459,359: Situational Recommendations and Control
- Angelos Angelopoulos (Graduate Intern) — co-inventor on two patent applications: Allocating Computing Resources for a Vehicle Application; Evaluation System for Determining a Network Connectivity Performance Road Map
- Zohreh Aghababaeyan (Graduate Intern) — co-inventor on U.S. Patent No. 12,459,359: Situational Recommendations and Control

Mentored at Clemson University under Prof. Mashrur Chowdhury (co-mentor), 2018 – 2021:

- Zadid Khan (Ph.D., Electrical Engineering background) — co-authored 2 journal papers and 1 book chapter
- Weimin Jin (Ph.D., Civil Engineering background) — co-authored 1 conference paper on AV lane-changing
- Eshaa Deepak Sood (M.S., Computer Science) — co-authored technical report on pedestrian safety
- Hsien-Wen Deng (M.S., Computer Science) — co-authored on connected vehicle cloud computing

PROFESSIONAL SERVICE

Journal Peer Review

- IEEE Transactions on Intelligent Transportation Systems
- IEEE Transactions on Vehicular Technology
- IEEE Internet of Things Journal
- IEEE Sensors

- Transportation Research Record (TRR)
- Cyber-Physical Systems (Taylor & Francis)
- Journal of Real-Time Image Processing

Conference Review

- IEEE Intelligent Transportation Systems Conference (ITSC)
- Transportation Research Board (TRB) Annual Meeting
- IEEE Intelligent Vehicles Symposium (IV)
- Society of Automotive Engineering (SAE)

PROFESSIONAL MEMBERSHIPS

- Member, Institute of Electrical and Electronics Engineers (IEEE)
- Member, IEEE Intelligent Transportation Systems Society (ITSS)
- Member, Society of Automotive Engineers (SAE)
- Member, Transportation Research Board (TRB) Friends of AI and Advanced Computing Committee

VOLUNTEERING & LEADERSHIP

- **President**, IEEE Intelligent Transportation Systems Society (ITSS) Student Chapter, Clemson University (2019 – 2020)
- **Treasurer**, Bangladesh Student Association, Clemson University (2017 – 2018)

PROFESSIONAL CERTIFICATIONS

Self-Driving Car Engineer Nanodegree <i>Udacity</i>	<i>Jan 2019</i>
Deep Learning Specialization <i>Coursera / deeplearning.ai</i>	<i>Apr 2018</i>
Internet of Things (IoT) Engineer Nanodegree <i>Udacity</i>	<i>Jul 2020</i>